



Centre of Excellence
Biopharmaceutical
Technology

Dec 2025 8th - 10th DEC CBT COURSE SERIES

COURSE BROCHURE



Biopharma 4.0
Critical Quality Attributes
Scale-up Operations
Upstream Development
Bioprocessing
Downstream Purification
Continuous Bioprocessing
Commercial Manufacturing
Machine Learning
Artificial Intelligence

Immunogenicity & Safety
Design Of Experiments
Regulatory Frameworks
New Age Biotherapeutics
Spectroscopy In Characterization



Welcome to the Centre of Excellence for Biopharmaceutical Technology

The Centre of Excellence for Biopharmaceutical Technology (CBT) was established at IIT Delhi in 2015 by the Department of Biotechnology, Government of India, in recognition of the importance of Biotechnology for India, particularly that of producing affordable biotech therapeutics.

The vision of CBT is to deliver innovation in biopharmaceutical technology to effectively address the challenges faced by the Indian biotech industry and thereby assist in the “Make in India” initiative by making India the global hub of manufacturing economical, safe and efficacious therapeutics.

CBT aspires to achieve this tall objective by providing a foundation for scientific and technological development to create novel technologies, engage with the biotech industry to translate these into applications, and finally to offer short term training courses to industry, academia, and regulatory agencies to facilitate creation of an ecosystem that delivers affordable biotech therapeutics to India and to the world.

CBT is bringing forth a world class training program that brings together leaders from across the world to come together and share the best practices and cutting-edge technologies on a diverse set of topics:

Commercial Biopharma Manufacturing - Best Practices for Process and Facility Operation-I	Commercial Biopharma Manufacturing - Best Practices for Process and Facility Operation-II	Drug Development: Concepts and Frameworks
Regulatory Framework for Lifecycle Management of Biotherapeutic Products	Immunogenicity and other Safety Concerns related to Biotherapeutic Products	Design of Experiments for Bioprocess Experimentation
Development and Optimization of Upstream processes	Scale-up and commercial operations in upstream processes	New Age Biotherapeutics: CAR-T cells, ADCs & VLPs
Critical Quality Attributes of Biotherapeutic Products	Role of Spectroscopy in Characterization of Biotherapeutic Products	Downstream processing development for biotherapeutic purification
Continuous Bioprocessing: Design, Integration, Monitoring, and Control	Analogue: A path to computational drug design	Machine Learning Fundamentals for Bioprocess Engineers

Biopharma 4.0: Path Towards Next Generation Biomanufacturing



Anurag S Rathore
Coordinator, CBT, IIT Delhi



James Gomes
Co-Coordinator, CBT, IIT Delhi

Registration Link: <https://forms.gle/7n96SHoP8VHRPWZ46>

THANK YOU SPONSORS!



December 8, 2025

Venue: LHC 413.1

Module 1 Commercial Biopharma Manufacturing - Best Practices for Process and Facility Operation-I

Description This module provides a step-by-step understanding of how modern biopharma facilities are planned and operated for large-scale drug production. It introduces participants to the essentials of infrastructure planning, capacity modeling, and facility design, before moving into best practices for upstream manufacturing processes. The module also explores how digital technologies and automation are transforming operations. By the end of this session, participants will have a clear view of how to align process development with facility needs to ensure efficiency, scalability, and global readiness.

OUTLINE

08:30-09:00 Breakfast

09:00-10:00 Inauguration (*Rangan Banerjee, Director, IIT delhi, Neeraj Jain, Director Growth Operations, PATH, Narendra Chirmule, SymphonyTech Biologics, James Gomes, IIT Delhi & Anurag S. Rathore, IIT Delhi*) @ LHC 416

10:00-10:30 Basics of Strategic Infrastructure Planning (*Sanjay B. Shah, Xenpharm Services*)

10:30-11:00 Tea Break

11:00-11:30 Basics of Overall Facility Planning and Design
(*E.K. Lee, Hanyang University-ERICA, Korea (South)*)

11:30-12:30 Capacity Planning, Process Mass Balance and Operations Modelling
(*Sanjay B. Shah, Xenpharm Services*)

12:30-14:30 Lunch

13:00-14:00 Best Practices for Commercial Scale Upstream Processes
(*Sanjay B. Shah, Xenpharm Services*)

14:00-14:45 Technology Transfer from R&D to manufacturing
(*Dhananjay Patankar, Industry Expert*)

14:45-15:30 Process Automation, March Towards Digitalization-I
(*Sanjay B. Shah, Xenpharm Services*)

15:30-16:00 Tea Break

16:00-17:00 Process Automation, March Towards Digitalization-II
(*Sanjay B. Shah, Xenpharm Services*)

17:00-17:45 Process Development Best Practices for Large Scale Biopharma
(*E.K. Lee, Hanyang University-ERICA, Korea (South)*)

17:45-18:30 Discussion + Q&A

18:30-20:00 Dinner

Purpose

Human beings are fragile beings, especially in Developing Countries, therefore they need to be protected. Unfortunately not everyone is protected and safe.

Diseases have been striking the world since its beginning, and the human population keeps evolving to fight and eradicate them.

DCVMN International comes to make a difference in protecting people.

We advocate, we support and we connect vaccine manufacturers from developing countries, and organise top of the deck training programs to help upgrade knowledge and skill sets of our member companies on continuous basis so that there can be a consistent and sustainable supply of quality vaccines at affordable prices to accelerate and reach the goal of global vaccine equity!

We work together, we provide knowledge, we leverage commercial opportunities.

And we do that for a single reason: so people can be protected from preventable diseases.

We do everything to protect, and everything we do, we do to protect.

DCVMN. We Connect to Protect.

We connect to protect!



This identity manual is mandatory to use the DCVMN brand in all applications.

For procedural guidelines in specific cases and situations not exemplified in this manual, consult the DCVMN communications team: a.marini@dcvmn.net

The use of copies, modifications, publications or public displays of this material or any of its content is strictly prohibited.

© DCVMN 2022. All rights reserved.

December 9, 2025

Venue: LHC 413.1

Module 2 Commercial Biopharma Manufacturing - Best Practices for Process and Facility Operation-II

Description Building on the foundations of Module 1, this module focuses on ensuring quality, compliance, and sustainability in commercial manufacturing. It covers critical topics such as commissioning and validation, cleaning validation, and quality management systems, along with the integration of external partners and supply chains. Participants will also learn how Tech Ops drives risk management and supports sustainability initiatives that go beyond traditional EHS practices. By the end of this session, participants will gain a practical framework for linking process operations with strong quality systems and sustainable growth strategies.

OUTLINE

08:30-09:00 Breakfast

09:00-09:50 Tech Ops and Management of External Supply, Contract Manufacturers, Partners
(*Dhananjay Patankar, Industry Expert*)

09:50-10:30 Phase Appropriate Quality Systems (*Sanjay B. Shah, Xenpharm Services*)

10:30-11:00 Tea Break

11:00-12:00 Elements of Process and Cleaning Validation
(*E.K. Lee, Hanyang University-ERICA, Korea (South)*)

12:00-13:00 Elements of Contemporary Practices, Common Challenges and Pitfalls in CQV
(*Commissioning-Qualification-Validation*) (*Sanjay B. Shah, Xenpharm Services*)

13.00-14:00 Lunch

14:00-14:45 Elements of Contemporary Practices with Quality Management Systems
(*Ravindra Khare, SymphonyTech Biologics, India*)

14:45-15:30 Elements of Quality Risk Management, Role of Competent Tech Ops
(*Ravindra Khare, SymphonyTech Biologics, India*)

15:30-16:00 Tea Break

16:00-17:30 Tech Ops as Key Enabler for Sustainability Initiatives. Conserving Water and Energy, Reduce Waste (*Beyond Traditional EHS*) (*Dhananjay Patankar, Industry Expert*)

17:30-18:30 Discussion + Q&A

18:30-20:00 Dinner

December 8, 2025

Venue: LHC 413.2

Module 3 Drug Development: Concepts and Frameworks

Description This module introduces participants to the foundational concepts of drug development, spanning discovery to preclinical evaluation. It highlights the critical stages of process development and optimization, emphasizing the role of Design of Experiments (DOE) in improving efficiency and decision-making. Core areas such as early-stage pharmacology, toxicology, and biologics-specific regulatory frameworks are explored. Through case-based discussions, interactive workshops, and expert-led sessions, attendees will gain an integrated perspective on how scientific, statistical, and regulatory elements intersect to shape the drug development pipeline.

OUTLINE

08:30-09:00 Breakfast

09:00-10:00 Inauguration (*Rangan Banerjee, Director, IIT delhi, Neeraj Jain, Director Growth Operations, PATH, Narendra Chirmule, SymphonyTech Biologics, James Gomes, IIT Delhi & Anurag S. Rathore, IIT Delhi*) @ LHC 416

10:00-11:00 Introduction to Drug Development
(*Narendra Chirmule, SymphonyTech Biologics, USA*)

11:00-11:30 Tea Break

11:30-12:30 Introduction to Process Development (*Nikhil Kateja, Edna Biolabs*)

12:30-13:00 Discussion + Q&A

13:00-14:00 Lunch

14:00-15:00 Introduction to Early Drug Development and Pharmacology
(*Vibha Jawa, EpiVax Inc., USA*)

15:00-15:30 Tea Break

15:30-16:30 Statistics for Early Drug Development
(*Ravindra Khare, SymphonyTech Biologics, India*)

16:30-17:30 Retrospective Analysis of Biologic Regulations
(*Priya Kulkarni, Independent Scientific Consultant*)

17:30-18:00 Introduction to Toxicology (*Narendra Chirmule, SymphonyTech Biologics, USA*)

18:00-18:30 Discussion + Q&A

18:30-20:00 Dinner

December 9, 2025

Venue: LHC 413.2

Module 4 Regulatory Framework for Lifecycle Management of Biotherapeutic Products

Description This module provides an in-depth overview of the regulatory considerations that govern the lifecycle of biotherapeutic products. Covering the continuum from clinical development (Phase I–IV) through manufacturing, participants will learn how key decision points are influenced by Target Product Profiles (TPP) and Probability of Technical Success (PTS). Topics include regulatory landscapes, trial design, statistical rigor, risk–benefit assessment, and the integration of Real-World Evidence (RWE) into decision-making. The program equips attendees with practical insights into dossier preparation, analytical requirements, and lessons learned from failed submissions, ensuring a strong foundation in regulatory compliance and strategy for biologics.

OUTLINE

08.30-09.00 Breakfast

09.00-10:00 Artificial Intelligence and Machine Learning (AI/ML) for Biological and Other Products Regulated by CBER (*Priya Kulkarni, Independent Scientific Consultant*)

10:00-11:00 Statistical Considerations for Regulatory Compliance in Clinical Trials
(*Ravindra Khare, SymphonyTech Biologics, India*)

11:00-11:30 Tea Break

11:30-12:30 Pre-Clinical Studies; Data Requirement, PK, PD & Toxicological Studies, Animal Model Selection (*Narendra Chirmule, SymphonyTech Biologics, USA*)

12:30-13:00 Discussion + Q&A

13:00-14:00 Lunch

14:00-15:00 CMC development for novel biologics (*Dhananjay Patankar, Industry Expert*)

15:00-15:30 Tea Break

15:30-16:30 Salient points from the newly revised Similar Biologics Guidance
(*Arvind Kukrety, CDSCO*)

16:30-17:30 Quality Considerations: Analytical Methods, Product Characterization and Specifications Development (*Abhiram Arunkumar, Sanofi, USA*)

17:30-18:00 Failures in Regulatory Filings: Lessons Learned
(*Narendra Chirmule, SymphonyTech Biologics, USA*)

18:00-18:30 Discussion + Q&A

18:30-20:00 Dinner



WHAT WE DO

Immunogenicity Insights for Therapeutic Development.

- In Silico Assessment
 - Software Licensing
 - Fee for Service Reports
- Validating In Vitro Assays
- Preclinical & Clinical Strategy Consulting

Modalities:

- Peptide, protein, fusion protein, & scaffold protein therapeutics
- Monoclonal, bispecific, multi-specific, single domain antibodies & other antibody-based constructs
- Gene & cell therapies (i.e. CAR-Ts, vectors, payloads)
- Generic & biosimilar therapeutics

Prevent immunogenicity surprises in the clinic.

EpiVax leads the industry in immunogenicity assessment and sequence optimization for biologic therapeutics and vaccines.

Utilizing proprietary methods for T cell epitope identification and characterization, we employ a wide range of validated in silico and in vitro techniques that are accepted by global regulatory agencies. This comprehensive approach provides accurate and thorough insights into candidate immunogenicity, ensuring the effectiveness and safety of our clients' products.

13M+

sequences
analyzed

18

of the 20 top
pharmaceutical
companies served

500+

industry, agency,
& academic
collaborators

230+

peer-reviewed
publications



[EpiVax.com](https://epivax.com)



info@epivax.com

December 10, 2025

Venue: LHC 413.2

Module 5 Immunogenicity and other Safety Concerns related to Biotherapeutic Products

Description Attendees will gain a comprehensive understanding of immunogenicity in biologic drug development, exploring its mechanisms, impact on safety and efficacy, and strategies for prediction and mitigation. The course will cover immune responses to peptides, proteins, gene and cell therapies, and mRNA vaccines, with real-world case studies highlighting key challenges. Participants will also learn about global regulatory guidelines, preclinical and clinical testing methods, and the use of silico and AI-based tools for risk assessment. Special focus will be given to emerging approaches such as personalized medicine and biotechnology innovations that shape the future of immunogenicity management.

OUTLINE

08:30-09:00 Breakfast

09:00-10:00 Introduction to Immunogenicity in Therapeutics (*Vibha Jawa, EpiVax Inc., USA & Narendra Chirmule, SymphonyTech Biologics, USA*)

10:00-11:00 Immunogenicity Prediction, Tools and their Outputs pre-recorded followed by live demo (*Aimee Mattei, EpiVax Inc., USA & Vibha Jawa, EpiVax Inc., USA*)

11:00-11:30 Tea Break

11:30-12:30 Regulatory Landscape and Immunogenicity Testing
(*Narendra Chirmule, SymphonyTech Biologics, USA*)

12:30-13:00 Discussion + Q&A

13:00-14:00 Lunch

14:00-15:00 Advanced Therapies and Immunogenicity Risk
(*Vibha Jawa, EpiVax Inc., USA*)

15:00-15:30 Tea Break

15:30-16:15 Immunogenicity of Peptides and Protein Therapeutics: Pre-recorded presentation
Aimee Mattei (EpiVax Inc., USA)

16:15-17:00 Immunogenicity Risk Assessment of Monoclonal Antibody Therapeutics
(*Sudeep Sabde, United States Pharmacopeia, India Pvt. Ltd.*)

17:00-17:45 The Future of Immunogenicity and Personalized Medicine
(*Guilhem Richard, EpiVax Inc., USA & Vibha Jawa, EpiVax Inc., USA*)

17:45-18:15 Statistical Issues in Analyzing Data from Immunogenicity Assays
(*Ravindra Khare, SymphonyTech Biologics, India*)

18:15-18:30 Discussion + Q&A

18:30-20:00 Dinner

December 8, 2025

Venue: LHC 416

Module 6 Design of Experiments for Bioprocess Experimentation

Description Design of Experiments (DOE) and Quality by Design (QbD) are powerful methodologies for enhancing processes, products, and outcomes across diverse industries, particularly in biopharmaceuticals. This training is designed to deepen participants' understanding of the QbD framework, with a strong emphasis on regulatory compliance and the use of JMP software's integrated workflows. Through expert-led sessions, attendees will gain practical skills in risk assessment, experimental design, and process optimization. By applying these approaches, participants can drive higher product quality, improved process robustness, and greater operational efficiency.

OUTLINE	Module Instructor: <i>Muralidhara Anandamurthy, JMP</i>
08:30-09:00	Breakfast
09:00-10:00	Inauguration (<i>Rangan Banerjee, Director, IIT delhi, Neeraj Jain, Director Growth Operations, PATH, Narendra Chirmule, SymphonyTech Biologics, James Gomes, IIT Delhi & Anurag S. Rathore, IIT Delhi</i>) @ LHC 416
10:00-11:15	Introduction to DOE: Strategy of Experimentation, Terminologies in DOE, Stages in Experimental Discovery, Guidelines in Analysis, Sequential Experimentation, Full Factorial Designs: Screening Designs: Fractional Factorial, Plackett Burman and Main Effects Screening Designs: Cases with Hands-On Exercises.
11:15-11:45	Tea Break
11:45-13:00	Classical Optimization Designs: Response Surface Methodology (RSM), Central Composite (CCD) and Box Behnken Designs (BBD), Prediction and Optimization: Cases with Hands-On Exercises
13:00-14:00	Lunch
14:00-16:15	Overview of Classical Mixture Designs, Mixture RSM and Scheffe Cubic Model, Prediction and Optimization. Easy DOE: Cases with Hands-On Exercises
16:15-16:45	Tea Break
16:45-18:00	Introduction to Computer Generated Designs, Custom Design, Design with Constraints, Concept of Quality by Design (Qbd), Mixture with Constraints, Mixture with Process Factors, Definitive Screening Designs (DSD): Cases with Hands-On Exercises
18:00-18:30	Discussion + Q&A
18:30-20:00	Dinner



You have a vision. We have a mission.

Eppendorf bioprocess solutions — for all your bioprocess needs

The Eppendorf bioprocess portfolio offers scalable bioprocess systems, single-use bioreactors, software solutions, and services to efficiently develop processes and move them from R&D to production.

- > Parallel benchtop bioreactor systems
- > Autoclavable and SIP fermentors
- > Rigid-wall single-use bioreactors with 65 mL to 40 L working volume
- > Precise and intuitive process control
- > Comprehensive information management

Discover application solutions, expert knowledge, product information and service support for a range of areas:

- > Development of cell and gene therapies
- > Upstream bioprocess development for antibodies
- > Vaccines
- > Modern food
- > Microbial production

Contact us: bioprocess-info@eppendorf.com

www.eppendorf.com/bioprocess

Eppendorf® and the Eppendorf Brand Design are registered trademarks of Eppendorf SE, Germany.
All rights reserved, including graphics and images. Copyright ©2024 by Eppendorf SE.



December 9, 2025

Venue: LHC 416

Module 7 Development and Optimization of Upstream Processes

Description The course “Development and Optimization of Upstream Processes” explores key strategies for enhancing cell culture-based biotherapeutic production. Sessions cover media technologies, rational media development, and stable cell line generation, alongside in-depth discussions of glycosylation as a critical quality attribute. Case studies demonstrate targeted manipulation of glycosylation and control of charge heterogeneity using media additives, while tutorials on multivariate data analysis and experimental design highlight tools for process optimization. The program integrates scientific insights with practical applications, concluding with an industry demo of the parallel bioreactor system.

OUTLINE

08:30-09:00 Breakfast

09:00-09:30 Overview of Culture Media Technology
(*Tom Fletcher, Fujifilm Biosciences, Santa Ana, USA*)

09:30-10:30 Tutorial- Rational Media Development
(*Tom Fletcher, Fujifilm Biosciences, Santa Ana, USA*)

10:30-11:00 Tea Break

11:00-12:00 Tutorial-How to Create Stable Cell Lines for Production of Biotherapeutic Products
(*Sanjeev Gupta, Ipca Laboratories Ltd., Mumbai, India*)

12:00-13:00 Glycosylation in Biopharmaceuticals: Impact on Critical Quality Attributes and Case Studies in Therapeutic Development (*Seongkyu Yoon, UMass Lowell, USA*)

13:00-13:15 Discussion + Q&A

13:15-14:00 Lunch

14:00-14:45 Effects of Cell Culture Media on Therapeutic Protein Quality
(*Tom Fletcher, Fujifilm Biosciences, Santa Ana, USA*)

14:45-15:30 Multivariate Data Analysis and Experimental Design for Process Control & Optimization (*Seongkyu Yoon, UMass Lowell, USA*)

15:30-16:00 Tea Break

16:00-16:30 Case Study 1: Targeted Manipulation of Glycosylation for Monoclonal Antibodies Through N-Glycosylation Modulators (*Neha Dixit, IIT Delhi*)

16:30-17:00 Case Study 2: Charge Heterogeneity Control in Mammalian Cell Culture Through Media Additives (*Vrinda Kapoor, IIT Delhi*)

17:00-18:30 Industry Demo: Eppendorf Single use Portfolio (Single use vessel bundle 50 L and 1 L Capacity) (*Eppendorf Group*)

18:30-20:00 Dinner



ENGINEERING YOUR BIOPROCESS. AT EVERY SCALE

Fermex Solutions is a leading provider of advanced bioprocessing equipment, specializing in custom-engineered fermentation and bioreactor systems. With a deep understanding of biological processes and industrial automation, we empower our partners to achieve unparalleled efficiency and scalability.

Certified Excellence

- ISO 9001: 2015 Certified
- ASME Compliance
- CE Mark
- WHO-GMP



CUSTOM SCALABILITY

- Custom-Engineered Systems
- Wide Volume Range: 2 Liters to 50,000 Liters
- Integrated Solutions: Fermenters, Automation & Downstream processing Equipment

PARTNER WITH FERMEX. DEFINE THE
FUTURE OF BIOPROCESSING



+91 818-1818-529
sales@fermex.in
info@fermex.in

149, JLPL Industrial
Area, Sector-82, Mohali-
140308, Punjab, INDIA

Flat No 201, MBR Sai Ram Apartment,
Cross Lane of Road No 10&11,
Panchasheel Enclave, HT Lane, Old Alwal,
Hyderabad, Telangana, 500010

December 10, 2025

Venue: LHC 416

Module 8 Scale-Up and Commercial Operations in Upstream Processes

Description The course “Scale-Up and Commercial Operations in Upstream Processes” offers a focused overview of strategies and challenges in translating laboratory processes to commercial manufacturing. Topics span high-performing perfusion processes, scale-up principles, monitoring and control of bioreactor operations, and advanced biotherapeutics such as viral vectors, ADCs, and AAV-based gene therapies. Case studies—including microbial production of Ranibizumab Fab and perfusion process intensification—highlight practical applications, while sessions on modeling, control, and an industry demo of the parallel bioreactor system to connect theory with practice.

OUTLINE

08:30-09:00 Breakfast

09:00-10:15 Tutorial - Development of a High Performing Perfusion Process
(*Tom Fletcher, Fujifilm Biosciences, Santa Ana, USA*)

10:15-10:30 Discussion + Q&A

10:30-11:00 Tea Break

11:00-12:00 Tutorial- Scale-Up and Commercial Bioreactor Operations (Monitoring and Control)
(*Sanjeev Gupta, Ipca Laboratories Ltd., Mumbai, India*)

12:00-13:00 Scale-Up Strategies in Bioreactor Systems: Principles, Challenges, and Case Studies in Biopharmaceutical Production (*Seongkyu Yoon, UMass Lowell, USA*)

13:00-13:15 Discussion + Q&A

13:15-14:00 Lunch

14:00-15:00 Process Development for Advanced Biotherapeutics: Viral Vectors, Antibody–Drug Conjugates, and AAV-Based Gene Therapies
(*Abhiram Arunkumar, Sanofi, USA*)

15:00-15:45 Modelling and Control Strategies for Upstream Process Development
(*James Gomes, IIT Delhi*)

15:45-16:00 Tea Break

16:00-16:30 Case Study 1: Microbial Upstream Process Development for Monoclonal Antibody Fragments: Ranibizumab (*Sami Ullah Bhat, IIT Delhi*)

16:30-17:00 From Bench to Plant: kLa-Driven Scale-Up in Stirred-Tank Bioreactors
(*Abhishek Mule, Eppendorf Group*)

17:00-18:30 Industry Demo: Eppendorf Single use Portfolio (Single use vessel bundle 50 L and 1 L Capacity) (*Eppendorf Group*)

18:30-20:00 Dinner

Biopharma Solutions & Services

Powering the Next Generation of Therapeutics



Our Core Offerings



Serum-Free Media Solutions:

- Chemically defined, animal component-free media designed for CHO cell productivity and scalability.



Bioprocess Development Services:

- Cell Line Development
- Upstream Development
- Downstream Development



Multicompendial Chemicals:

- Integrate seamlessly into every stage of your biopharma workflow ensuring consistency, compliance, and quality.



Analytical Services:

- Protein Quality Control
- Stoichiometry of protein complexes
- Plasma Proteomics
- Glycan Analysis



Proven expertise in media engineering & bioprocess design



Regulatory-ready methods for biosimilars



Streamlined R&D-to-production transitions



Trusted partner in global biologics development

Why HiMedia?

✉ atc@himedialabs.com



+91-22-6147 1919 / 6116 9797 / 6903 4800
Fax: +91-22-6147 1920



HiMedia Laboratories™
www.himedialabs.com

December 8, 2025

Venue: LHC 413.3

Module 9 New Age Biotherapeutics: CAR-T cells, ADCs & VLPs

Description This course highlights the next wave of innovative biotherapeutics - antibody drug conjugates, virus-like particles (VLPs), and CAR-T therapies—providing insights into their analytical challenges, translational opportunities, and manufacturing considerations. Designed to serve both industry professionals aiming to accelerate development pipelines and academic researchers seeking to expand expertise in cutting-edge modalities, the program offers a unique platform to amalgamate scientific knowledge with translational impact. This comprehensive program bridges scientific fundamentals with real-world applications in the evolving field of biotherapeutics.

OUTLINE

08:30-09:00 Breakfast

09:00-10:00 Inauguration (*Rangan Banerjee, Director, IIT delhi, Neeraj Jain, Director Growth Operations, PATH, Narendra Chirmule, SymphonyTech Biologics, James Gomes, IIT Delhi & Anurag S. Rathore, IIT Delhi*) @ LHC 416

10:00-11:15 CAR-T Cells (*Vibha Jawa, EpiVax Inc., USA*)

11:15-11:45 Tea Break

11:45-13:00 Next-Generation Strategies in The Production, Purification & Characterization of Antibody–Drug Conjugates (*E.K. Lee, Hanyang University-ERICA, Korea (South)*)

13:00-14:00 Lunch

14:00-15:15 Tutorial -Introduction on Virus Like Particles (VLPs): Manufacturing and Their Analytical Characterization) (*Nico Lingg, BOKU University, Austria*)

15:15-15:35 Case Study 1: Upstream Process Development of HPV- VLPs (*Meenal, IIT Delhi*)

15:35-15:55 Case Study 2: Purification Strategies for HPV- VLPs (*Rashmi Sharma, IIT Delhi*)

15:55-16:15 Case Study 3: Analytical Characterization of HPV-VLPs (*Vineela P., IIT Delhi*)

16:15-16:45 Tea Break

16:45-18:30 Industry Demo: Duetta & ViewSizer 3000 (*Horiba*)

18:30-20:00 Dinner

Smart science. Seamless solutions

Raman microscopes & spectrometers



XploRA PLUS



LabRAM Odyssey



LabRAM Soleil



Raman Plate Reader



Particle Characterization Analyzes



LA-350



LA-960V2



SZ-100V2



ViewSizer 3000



Fluorescence Spectrometers



Aqualog



FluoroMax Plus



Duetta



Veloci



December 9, 2025

Venue: LHC 413.3

Module 10 Critical Quality Attributes of Biotherapeutic Products

Description This intensive one-day course provides participants with a structured understanding of Critical Quality Attributes (CQAs) in biotherapeutics and the modern analytical strategies used to define, measure, and control them across the product lifecycle. This module also covers the principles of aQbD, regulatory expectations, and how risk-based analytical methods ensure CQA control throughout the lifecycle. With a balance of conceptual frameworks, regulatory context, and case-studies, the course is designed to bridge industry needs for robust quality systems with academic insights into cutting-edge analytical science.

OUTLINE

08:30-09:00 Breakfast

09:00-10:30 Critical Quality Attributes (CQAs) for Therapeutic Monoclonal Antibodies
(*Abhiram Arunkumar, Sanofi, USA*)

10:30-11:00 Tea Break

11:00-12:00 Analytical and Functional Methods for CQA Characterization
(*Srishti Joshi, Mahindra University, Hyderabad, India*)

12:00-13:00 Integrating Mass Spectrometry Workflows for The Identification of Critical Quality Attributes (*Sunil Kumar, University of Pavia, Italy*)

13:00-14:00 Lunch

14:00-15:00 From Case Studies to Compliance: The Role of Multi-Attribute Methods in Modern Biologics Characterization (*Srishti Joshi, Mahindra University, Hyderabad, India*)

15:00-15:45 Case Study 1: Decoding The Glycan Fingerprint: MS-Based Glycosylation Profiling of Therapeutic Proteins (*Sunil Kumar, University of Pavia, Italy*)

15:45-16:15 Case study 2: Characterization of critical quality attributes through 2-dimensional liquid chromatography (*Kunal Krishna, IIT Delhi*)

16:15-16:45 Tea Break

16:45-18:30 Industry Demo: Duetta & ViewSizer 3000 (*Horiba*)

18:30-20:00 Dinner

December 10, 2025

Venue: LHC 413.3

Module 11 Role of Spectroscopy in Characterization of Biotherapeutic Products

Description This module is focused on exploring the emerging role of spectroscopy to address the challenges of biotherapeutic characterization. This course will cover fundamentals and applications of spectroscopy in transforming quality assessment into biopharma. The course blends real-world case studies to gain insights into applications of spectroscopy in biopharma.

OUTLINE

08:30-09:00 Breakfast

09:00-10:30 Spectroscopic Methods: Working Principles, Measurement Parameters, and Critical Considerations-I (*Nico Lingg, BOKU University, Austria*)

10:30-11:00 Tea Break

11:00-12:00 Introduction to A-TEEM spectroscopy as a technique in Biopharma industry (*HC Sudeeksha, Horiba*)

12:00-13:00 From Lab to Clinic: Circular Dichroism for Next-Generation Diagnostics and Biotherapeutics (*Chinmoyee Maharana, Jammu University*)

13:00-14:00 Lunch

14:00-14:45 Role of Spectroscopy for Process Control (*Anurag S. Rathore, IIT Delhi*)

14:45-15:15 Case Study 1: Near-Infrared (NIR) Spectroscopy based therapeutic Monoclonal Antibody Concentration and Excipient Concentration Quantification (*Naveen G Jesubalan, IIT Delhi*)

15:15-15:45 Case Study 2: Raman spectroscopy-based Characterization of Charge variants of monoclonal antibodies. (*Nitika, IIT Delhi*)

15:45-16:15 Case study 3: Fourier Transform Infrared (FTIR) Spectroscopy based Counterfeit Detection of Biotherapeutics (*Vineela P., IIT Delhi*)

16:15-16:45 Tea Break

16:45-18:30 Industry Demo: Horiba: Duetta & ViewSizer 3000 (*Horiba*)

18:30-20:00 Dinner

Engage with Ease: Connect, Request, and Provide Feedback

Our customers asked, we listened.

Pall Life Sciences and Cytiva have united under the Cytiva name — forming an even stronger global provider of technologies and services that advances and accelerates the development, manufacture, and delivery of therapeutics.

Explore how we can empower you

- Bioprocessing Solutions
- Cell and Gene Therapies
- Discovery/Research
- Laboratory Products
- Process Development
- Training and Support



December 9, 2025

Venue: LHC 418

Module 12 Downstream Processing Development for Biotherapeutic Purification

Description The course on “Advanced manufacturing approaches for downstream bioprocessing” offers a comprehensive exploration of downstream processing for biotherapeutics, focusing on the design and industrial application of protein purification strategies. It begins with an overview of protein purification and key unit operations, followed by detailed discussions on filtration fundamentals, chromatography principles, and scale-up approaches. Participants will examine advanced purification methods for emerging modalities such as bispecific antibodies and novel protein formats, along with the associated development challenges and solutions. Expert-led sessions include alternative purification strategies, academic and industry case studies, and practical guidance for initiating purification workflows for new proteins. The program emphasizes translating lab-scale methods into robust, scalable manufacturing processes, with insights into development and quality considerations.

OUTLINE

08:30-09:00 Breakfast

09:00-09:45 How To Develop a Purification Process for a mAb Product – Choice of Unit Operations, Integration, Optimization (*Nico Lingg, BOKU University, Austria*)

09:45-10:45 Development and Scale-Up of a Commercial Chromatography Step
(*Rahul Bhambure, National Chemical Laboratory, Pune*)

10:45-11:00 Tea Break

11:00-12:00 Development and Scale-Up of a Commercial UF/DF Step
(*Abhiram Arunkumar, Sanofi, USA*)

12:00-13:00 Use of Process Modelling for Optimization and Characterization of Downstream
(*Rahul Bhambure, National Chemical Laboratory, Pune*)

13:00-14:00 Lunch

14:00-15:00 Driving Efficiency in Biotherapeutic Purification: Intensified Downstream Solutions
(*Soumen Dey, Cytiva*)

15:00-15:30 Case Study 1: Downstream Processing of Novel Nanobodies
(*Subhankar Metya, IIT Delhi*)

15:30-16:15 Case Study 2: Simultaneous Removal of Aggregates and Charge Variant of mAbs
Using a Single Step Chromatography (*Rashmi Sharma, IIT Delhi*)

16:15-16:30 Tea Break

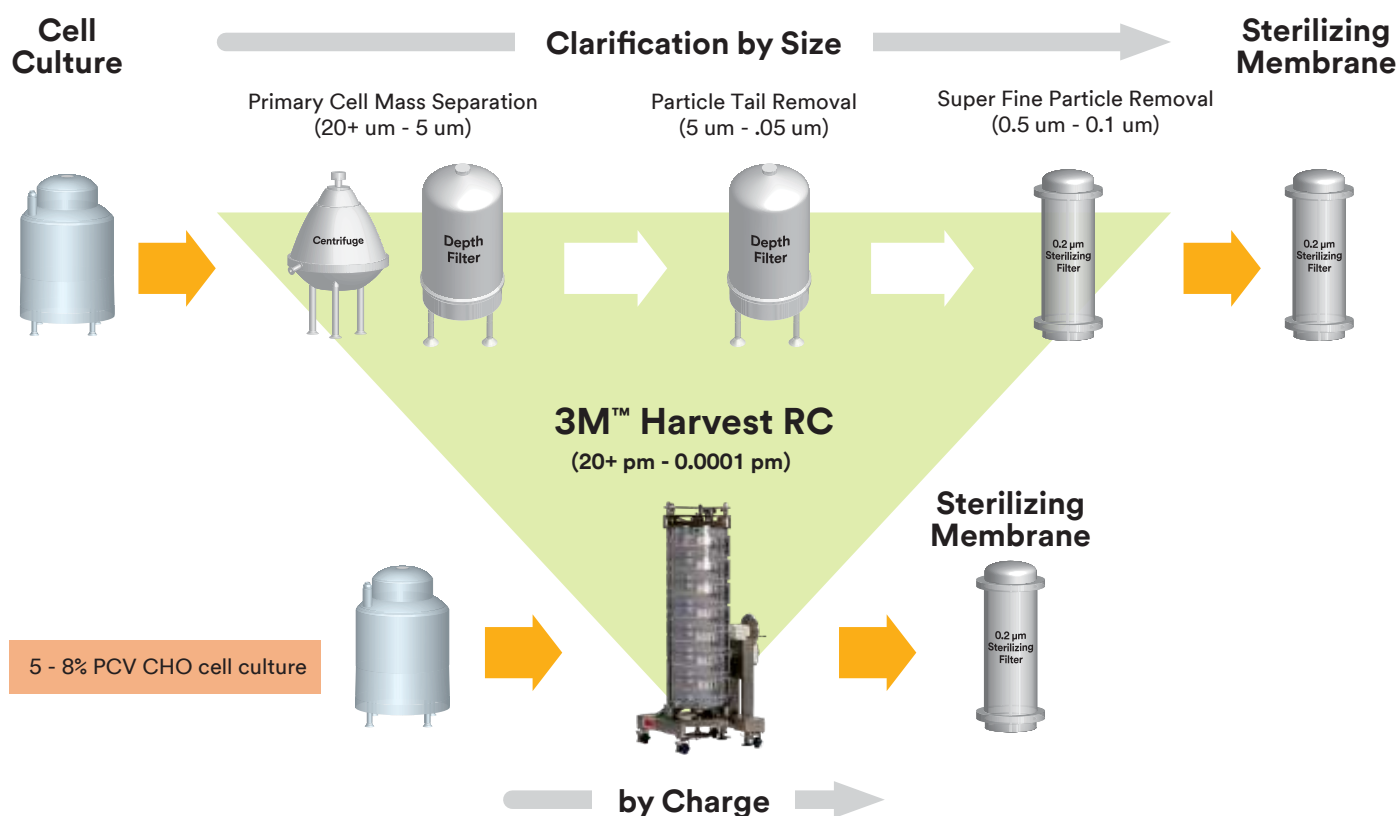
16:30-18:30 Industry Demo 1: Chromatographic Clarification of Cell Culture harvest (*3M*)
Industry Demo 2: *Cytiva*

18:30-20:00 Dinner



Single-stage harvest and clarification

Simplify three stages into a single stage



3M™ Harvest RC

- High mAb product recovery (>95%)
- Predictable Scalability
- Consistent effluent quality
- Turbidity reduction (<15 NTU)
- DNA reduction (<500 ppb)
- Minimal cell shear
- 0.2 μm sterile filter protection



December 10, 2025

Venue: LHC 418

Module 13 Continuous Bioprocessing: Design, Integration, Monitoring, and Control

Description This course on Continuous Bioprocessing: Design, Integration, Monitoring, and Control provides a strong foundation and practical insights into continuous downstream processing for biotherapeutics. With the rising need for cost-effective, scalable, and efficient platforms, continuous purification is highlighted as a transformative approach to improve process robustness and product quality. The program covers fundamentals of continuous bioprocessing and enabling technologies such as chromatography, refolding, and precipitation. Participants will learn how to integrate these steps into an end-to-end continuous manufacturing train and explore strategies for scale-up from lab to commercial production. Sessions will also focus on economics, cost reduction, accelerated timelines, and CQA-based regulatory compliance. Case studies from academia and industry will provide real-world insights into implementation challenges, solutions, and lessons learned.

OUTLINE

08:30-09:00 Breakfast

09:00-10:00 Continuous Bioprocessing for Process Intensification (*Anurag S. Rathore, IIT Delhi*)

10:00-11:00 How To Design a Continuous Chromatography Step
(*Maarten Pennings, Pennings Bioprocess Consultancy, The Netherlands*)

11:00-11:15 Tea Break

11:15-12:00 How To Design a Continuous UF/ DF Step (*Nico Lingg, BOKU University, Austria*)

12:00-13:00 How To Integrate Individual Operations to Create a Continuous Process
(*Maarten Pennings, Pennings Bioprocess Consultancy, The Netherlands*)

13:00-13:45 Lunch

13:45-14:30 How To Perform Process Monitoring and Control in Continuous Processing
(*Maarten Pennings, Pennings Bioprocess Consultancy, The Netherlands*)

14:30-15:15 Continuous Process Economics and Impact on Sustainability
(*E.K. Lee, Hanyang University-ERICA, Korea (South)*)

15:15-16:15 Understanding the Impact of Nuclease Elimination in AAV Manufacturing Process Using 3M™ Chromatographic Clarifier (*Vishal Vijayan, 3M*)

16:15-16:30 Tea Break

16:30-18:30 Industry Demo: Chromatographic Clarification of Cell Culture harvest (*3M*)
Industry Demo 2: *Cytiva*

18:30-20:00 Dinner

December 8, 2025

Venue: LHC 413.4

Module 14 Analogue: A Path to Computational Drug Design

Description Join our one-day course introducing the essentials of computational drug discovery through QSAR, docking, and molecular dynamics (MD). Learn how AI-driven generative and predictive models accelerate molecule design. Explore real-world case studies on chalcone tubulin interactions. hands-on experience with the Analogue platform for practical applications. Part of the Analogue Course Series, designed to train future innovators in computational biology.

OUTLINE **Module Instructor:** *Avinash Mishra*

08:30-09:00 **Breakfast**

09:00-10:00 Inauguration (*Rangan Banerjee, Director, IIT delhi, Neeraj Jain, Director Growth Operations, PATH, Narendra Chirmule, SymphonyTech Biologics, James Gomes, IIT Delhi & Anurag S. Rathore, IIT Delhi*) @ LHC 416

10:00-10:30 Application of Computational Biology for Drug Discovery

10:30-10:45 **Tea Break**

10:45-11:15 QSAR Unplugged: Turning Molecules into Mathematical Insights

11:15-11:45 Molecule Maker: Generative AI in Drug Discovery

11:45-12:15 Seeing The Activity: Predictive AI for Smarter Drug Design

12:15-13:00 Dynamics of Drug Design – Part I

13:00-14:00 **Lunch**

14:00-14:45 Advanced Molecular Dynamics – Part II

14:45-15:30 Case Study 1: Benchmark Study for QSAR Modeling of Chalcone Derivatives for Targeting Tubulin Protein Using ANALOGUE Platform

15:30-16:00 **Tea Break**

16:00-17:00 Case Study 2: Benchmark Study for Docking + Molecular Dynamics (MD) for Chalcone-Tubulin

17:00-18:30 Case Studies + Analogue Demo

18:30-20:00 **Dinner**



About

ACS Student chapter

ACS student chapters are for scholars with an interest in the chemical sciences. Members participate in a wide range of programs and activities that enhance their college experience and prepare them for successful careers.

Discover the value of ACS membership

1. FREE LinkedIn Learning Premium Courses worth ₹ 20,000
2. Weekly issues of Chemical & Engineering News (C&EN) digital edition
3. Weekly brainstorming and journal club
4. Industry insights



 acsstudentchapter.india@gmail.com

 ACS International Student Chapter: IIT Delhi India, IIT Delhi

Why Join ACS?

Join the American Chemical Society and unlock numerous member benefits tailored to support your professional growth and networking opportunities in the field of chemistry.

Membership Pricing

B.Tech : ~ 800 INR (Annual)

M.Tech/PhD : ~ 1,700 INR (Annual)

Additional perks when registering through us

Journal Club sessions,
Exclusive chapter webinars,
Complementary goodies.

Become a member today and enhance your career in chemistry.



Join the Whatsapp community

December 9, 2025

LHC 413.4

Module 15 Machine Learning Fundamentals for Bioprocess Engineers

Description The course is designed for bioprocess engineers, life science researchers, data enthusiasts in biotechnology, and professionals interested in applying ML and AI techniques to process optimization and decision-making in bioprocess industries. By the end of this course, participants will be able to understand the fundamentals of data handling, preprocessing, and feature engineering for bioprocess datasets, apply good practices in managing data quality, avoiding biases, and ensuring reproducibility, build and evaluate ML models using classical algorithms and interpret their results, gain practical knowledge of deep learning algorithms and their applications in process modeling and control, utilize explainable AI techniques to make models transparent and interpretable, and understand reinforcement learning concepts while evaluating its role in adaptive process optimization.

OUTLINE

08:30-09:00 Breakfast

09:00-09:50 Introduction: Data Handling, Feature Selection and Optimization Techniques
(*Manojkumar Ramteke, IIT Delhi*)

09:50-10:40 Supervised Learning: Algorithms, Evaluation Metrics and Case Studies
(*Om Prakash, IIT Delhi*)

10:40-11:10 Tea Break

11:10-12:10 Unsupervised Learning: Clustering, Dimensionality Reduction, Association Rules
(*Om Prakash, IIT Delhi*)

12:10-13:10 Introduction To Neural Networks: DNN / CNN / RNN/ LSTM
(*Manojkumar Ramteke, IIT Delhi*)

13:10-14:00 Lunch

14:00-15:00 Graph Neural Network: Fundamentals and Case Study (*Om Prakash, IIT Delhi*)

15:00-15:45 Physics Informed Neural Network: Basics and Case Study
(*Md Nasre Alam, IIT Delhi*)

15:45-16:15 Tea Break

16:15-17:00 Q-Table based Reinforcement Learning - Basics and Case Study
(*Naveen G Jesubalan, IIT Delhi*)

17:00-17:45 Explainable AI: LIME and SHAP (*Ramya Lakshmi, IIT Delhi*)

17:45-18:30 From Data to Decisions: Best Practices for PAT Tool Development Success
(*Seongkyu Yoon, UMass Lowell, USA*)

18:30-20:00 Dinner

We believe great inventions make real world impact

At TCS Research, we believe in the power of inspiration and invention to build greater futures. Our world-class researchers apply scientific rigor and a collaborative mindset to solve problems faced by industry and society.

Make #TheBigMove Towards [Building on belief](#)

Reach out to us careers.research@tcs.com



Scan to know more

Converse with us on

 @Tata Consultancy Services - Research

 @tcsresearch

with #TCSResearch

[A boy reads one of the 650,000+ books made accessible to people with print disability under the Sugamya Pustakalaya program, a joint effort between TCS Research, Daisy Forum of India, and National Institute for Empowerment of Persons with Visual Disabilities, Government of India.]

December 10, 2025

LHC 413.4

Module 16 Biopharma 4.0: Path Towards Next Generation Biomanufacturing

Description Biopharma 4.0: Path Towards Next Generation Biomanufacturing introduces participants to the digital transformation of the biopharmaceutical industry, integrating Industry 4.0 concepts such as automation, AI/ML, IoT, and digital twins with modern bioprocessing. The course covers continuous manufacturing, advanced process monitoring, and data-driven decision-making to improve efficiency, quality, and scalability of biologics production. Learners will gain insights into regulatory considerations, real-time release testing, and strategies for adopting smart manufacturing practices. Through case studies and practical examples, they will understand how emerging technologies reshaping drug development and production are. The key takeaway is the ability to apply Biopharma 4.0 tools and approaches to drive innovation, compliance, and sustainability in next generation biomanufacturing.

OUTLINE

08:30-09:00 Breakfast

09:00-09:45 Current trends in Biopharma 4.0 implementation (*Vinesh B. Yezhuvath, TCS*)

09:45-10:30 Biopharma 4.0 Enablers: Part 1 Modeling, Optimization & Control
(*Venkat Runkana, TCS*)

10:30-11:00 Tea Break

11:00-11:45 Biopharma 4.0 Enablers: Part 2 Technology Transfer (*Vinesh B. Yezhuvath, TCS*)

11:45-12:30 Role of Technology in Innovation and Growth: Insights (*Sai Prasad, TCS*)

12:30-13:15 Role of AI -ML in Next Generation Biopharmaceutical Manufacturing
(*Anurag S. Rathore, IIT Delhi*)

13:15-14:00 Lunch

14:00-14:45 Case study 1: Bioreactor Modeling & Digital Twin (*V. Sudheendra Buddhiraju, TCS*)

14:45-15:30 Case study 2: Multi-column Chromatography Digital Twin (*P. Karundev, TCS*)

15:30-16:15 Soft-sensor Applications in Process Monitoring (*Venkat Runkana, TCS*)

16:15-16:45 Tea Break

16:45-17:45 Multi-objective Optimization (*Manojkumar Ramteke, IIT Delhi*)

17:45-18:30 Case Study 3: Multi-objective optimization (*Naveen G Jesubalan, IIT Delhi*)

18:30-20:00 Dinner

OUR SPEAKERS



ANANDAMURTHY MURALIDHARA is an accomplished analytics and data science leader with over 23 years of experience spanning industry and academia. He currently serves as Global team member at the JMP Division of SAS Institute, driving adoption of the JMP statistical discovery platform in India and promoting statistical thinking, design of experiments, and machine learning across sectors. Previously, he led analytics teams at Danske Bank and Target Corporation, shaping data strategies, governance, and high-impact marketing analytics functions. He began his career at Genpact, applying statistical models and Lean principles to business problems. A visiting faculty at premier institutes like IISc, IITs, and IIMs, he teaches statistics, DOE, and Six Sigma. Holding a PhD, MBA, and B.Tech, he is also co-author of Machine Learning for Business Analytics.



ARUNKUMAR ABHIRAM is currently a Senior Director – Cluster Head for Rare Disease products within Sanofi's Manufacturing & Supply Organization in Boston, USA. In his current role, he leads end-to-end lifecycle management of Chemistry, Manufacturing and Controls (CMC) aspects of several recombinant protein therapeutics in the Sanofi portfolio. Prior to Sanofi, Dr. Arunkumar has worked in different leadership roles at Voyager Therapeutics, Amgen and Bristol-Myers Squibb in different functions – Process Development, Manufacturing Sciences and Single Use Technologies – for both Biologics and AAV Viral Vectors. Dr. Arunkumar has a PhD from the University of Wisconsin-Madison, where he worked on electrically charged ultrafiltration membranes for bioprocessing. He has a bachelor's degree in engineering from the Institute of Chemical Technology, Mumbai. Dr. Arunkumar is passionate about CMC and process improvements for biologics including integrated continuous bioprocessing.



BHAMBURE RAHUL is currently leading the Bioprocess Technology group in Chemical Engineering and Process Development Division of CSIR - National Chemical Laboratory, Pune. My area of research covers both experimental and theoretical grounds applied mainly to the field of process and product development for biopharmaceuticals. For the last ten years, I have been working in bioprocess development for recombinant protein and nucleic acid therapeutics. My future research interest will be more focused on applied protein biophysics and biochemical engineering for developing novel processes at various stages of biopharmaceutical manufacturing like upstream, downstream and formulation.



BUDDHIRAJU V. SUDHEENDRA is Senior Scientist at TCS Research. He has a Ph.D. from IIT Kanpur in chemical engineering. He has more than 15 years of experience in Process modeling and simulation, Nanoparticle synthesis and growth, Ultrafine grinding, Functional nanofibers, Energy storage materials, CFD and DEM approaches for process modeling and drug delivery systems.



CHIRMULE NARENDRA is a co-founder of Symphony Technologies, a data analytics company. He currently teaches drug development at UPenn, and Indian Institute of Technology, Mumbai and Delhi. As former Head of R&D at Biocon (Bangalore), and in leadership positions at Amgen (Thousand Oaks, CA) and Merck Vaccines (West Point, PA), he has contributed to the clinical development of vaccines, biologics, and cell-and-gene therapies. During his academic career, he has worked on the development of a leprosy vaccine, the pathogenesis of AIDS (Cornell, New York, NY), and gene therapy for several rare diseases (UPenn, Philadelphia, PA). Dr Chirmule is on the NIH advisory committee for HIV vaccines, and on department of biotechnology grant scientific review committee. His TEDx talk and book "Good Genes Gone Bad" by Penguin Press, describe lessons learned from colossal failures in drug development. In 2025, he has edited a textbook "Approved: Life Cycle of Drug Development" by Springer.



DEY SOUMEN leads the Protein Research Hardware business for Cytiva in South Asia, where he drives product vision, strategy, & roadmap to deliver holistic solutions aligned with evolving market needs. With over 14 years of experience in bioprocessing & life sciences, Soumen has been instrumental in enabling speed, efficiency, & capacity in research & manufacturing workflows across the region. He holds an M.Tech degree from the Institute of Chemical Technology (formerly UDCT). Soumen began his career with a leading Indian-global biopharmaceutical manufacturer before joining Cytiva (then GE Healthcare), where he has held diverse roles spanning Applications, Commercial, & Product Management. Passionate about building trust-based partnerships, Soumen has forged strategic alliances with industries & incubation centers. Through his consulting efforts, he actively drives innovation & promotes operational excellence in protein research & biomanufacturing.



FLETCHER TOM is Director of Strategic Programs at Fujifilm Biosciences, a company that specializes in the development, manufacture and global supply of cell culture media used for producing therapies and vaccines. Tom leads the company's efforts to provide relevant custom solutions to the various and frequent bioprocessing challenges faced by biopharmaceutical and other cell-culture-based manufacturers. He has over 30 years of experience in developing cell culture media for large-scale production processes. Tom has previous experience in protein chemistry research at the University of California, Irvine School of Medicine and working at Becton Dickinson & Co.



GOMES JAMES received his bachelor's degree in chemical engineering from Jadavpur University and M.S. and Ph.D. from Tulane University, New Orleans, USA. As a graduate student at Tulane, he received several awards including the prestigious Sigma Xi research award. He worked as Senior Process Engineer for Process Services Inc., Baton Rouge and as Executive Engineer for L&T, Powai, Mumbai, before joining IIT Delhi. His current research interest is in Systems and Network Biology. Much of Prof. Gomes' research has been in the area of nonlinear dynamics and control of bioprocesses. His group also focusses on the application of mathematical and computational methods to study disease networks, understand disease progression and determine new drug targets, followed by directed experiments. Particular areas include neurodegenerative diseases and virus-host interaction networks. He also has interest in emergent properties arising in complex systems and a growing interest in the brain, cognition and consciousness problem.



GUPTA SANJEEV has obtained Ph.D. in Cell Line Engineering and M.Sc. in Applied Microbiology and Biotechnology. He also received Honorary Professorship award from Amity University, MH (Feb. 2024). At present working as Sr. Vice President and BU Head-Biosimilar Division (R&D, Mfg. and Business Dev.), Ipca Laboratories Ltd., Mumbai, India. Dr. Gupta carries around 25 years of industrial experience and has been working since 2000 on the development of various "Biosimilars" including monoclonal antibodies. His core expertise lies in Molecular Biology, Cell Line, Process Development, Regulatory, Tech Transfer and Scale-up in the GMP for Clinical and Commercial manufacturing. He is actively involved in dossier submission, approval and interaction with various regulatory agencies such as EMA, MHRA-UK, CDSCO and US-FDA for Biosimilar launch in the overseas markets. In addition, he is closely working with Indian regulatory bodies (RCGM/CDSCO) and Govt. agencies (BIRAC) for Guidelines reforms and project selection/approvals, respectively. Till date, he developed and contributed for over Fifteen (15) recombinant molecules, of which Nine (09) biosimilars are already launched in the various markets and the rest are under development or clinical phase, and are expected to be launched in near future. He delivered talks on Biosimilars in various national and international conferences organized by IBC, Terrapin, BPI, BioTrain, IMAPAC, CphI, Biopharma, Informa and UBM. He also chaired/Moderated various sessions in the national and international forums for conducting the biological conferences. He has published several Research articles (15), Book chapters (06), Magazines (06) and Patent (01) in lieu of the Biosimilar development. During his professional tenure, he worked at various position in reputed Indian biopharmaceuticals industries such as Zydus Ahmedabad, Wockhardt-Aurangabad, Panacea-New Delhi, Intas-Ahmedabad and at present with Ipca Laboratories, Mumbai.



H C SUDEEKSHA is an Applications Scientist and Manager – Applications and Business Support at HORIBA India Pvt Ltd, Bangalore. He has a Bachelor of Science Degree in Biotechnology from St. Aloysius College, Mangalore and Master of Science degree in General Chemistry from National Institute of Technology, Karnataka. He graduated with a Ph.D. in Physical Chemistry at Indian Institute of Science, Bangalore in 2020 and has been working with HORIBA India Pvt Ltd. ever since. His research areas of interest are optical spectroscopy, specifically Raman and fluorescence spectroscopy, time-resolved spectroscopy, computational chemistry, and multivariate analysis. He is a recipient of the UGC Junior, and Senior Research Fellowships and he was a Summer Research Fellow of the Indian Academy of Sciences. In his capacity as the Manager of Applications and Business support, he is constantly involved in engaging with the key stakeholders of the Biopharma and other Life Sciences industry and promotion of new technologies for R&D and QC/QA applications.



JAIN NEERAJ drives growth strategy, country operations, and government engagement for PATH across Asia, the Middle East, and Europe. Under his leadership, PATH has significantly expanded its global health impact by partnering with innovators and governments to address critical public health challenges. With over 35 years of experience in market expansion, change management, and strategic development, Neeraj has driven large-scale initiatives across public health, climate and health, digital innovation, sanitation, and child development. Neeraj has held leadership roles at PATH, WaterAid India, the Children's Investment Fund Foundation, Vestergaard Frandsen, and the India Sanitation Coalition, delivering transformative impact across Asia and Europe. He actively contributes to public health through roles on national committees such as the Confederation of Indian Industry (CII)'s Nutrition Committee and serves on the boards of Social Alpha, Digital Green, Arogya World, and NATHEALTH. Neeraj holds an MBA from the Faculty of Management Studies (FMS), University of Delhi, and a bachelor's degree in business studies from Hong Kong Polytechnic University.



JAWA VIBHA is Chief Scientific Officer of EpiVax, where she leads scientific strategy, guiding research efforts and helping shape the company's approach to new therapies and technologies. Previously, she held the position of Executive Director in the Translational Medicine organization at Bristol Myers Squibb. In this role, she led biotherapeutic and cell /gene therapy bioanalytical (BA) function, supporting PK and immunogenicity related preclinical and clinical related support. She provided strategic and scientific oversight for BMS developmental portfolio. Vibha was at Merck for 4 years where she led the Predictive and Clinical Immunogenicity group and at Amgen for 14 years supporting Discovery to Development for biotherapeutics. Vibha has 20+ years of experience in diverse fields of biologics, vaccine development and gene therapy with successful support of 20 + IND, BLA and MAA filings. Vibha is a recognized leader in Bioanalysis and Immunogenicity with 75+ peer-reviewed publications and serves as a Reviewer and Editor for The AAPS Journal and J. Pharm Sci. She is an active member of multiple scientific societies and consortiums (IQ, SC space Consortium and EIP). Within AAPS, she is Track Chair of Land O Lakes Bioanalysis Meeting, Chair of the Cell and Gene therapy Bioanalysis and Biomarker working group, Steering

Committee member of the Therapeutic Product Immunogenicity Community, past chair of Immunogenicity Risk Assessment and Mitigation Community and leads the IQ Consortium for Cell/Viral/Gene therapies. She was recognized as the AAPS Fellow 2022. Vibha is the President of Steampark a nonprofit promoting STEM based learning in underserved communities. She also likes to mentor high school students on STEM related projects and early career scientists in her free time.



JOSHI SRISHTI is an Assistant Professor at Centre for Life Sciences in Mahindra University (MU). Her research expertise lies in analytical characterization of critical quality attributes of biopharmaceuticals, particularly biosimilars, through biophysical techniques for wide applications. She holds a MSc in Crop biotechnology and Entrepreneurship from University of Nottingham, UK and a PhD in molecular biology and biochemistry from Massey University, NZ with post doc experience from Massey University, Korea University, and IIT Delhi. Prior to MU she served as a Post Doc and researcher under the tutelage of Prof. Anurag Singh Rathore at the DBT Centre of Excellence for Biopharmaceutical Technology, at IIT Delhi for 7 years. In addition to research, she has served in administrative roles as the Coordinator of Bioprocessing Society of India (BPI; 2021-23) and Coordinator of International Programs, IITD (2023-24). At MU, in addition to research, she also serves as Program Coordinator for MSc by Research in Biotherapeutics.



KATEJA NIKHIL is the Co-Founder and CEO of Edna Biolabs Private Limited, a young start-up company involved in the manufacturing of specialty reagent and kits for research and In-vitro diagnostics. He has done his bachelor's in biotechnology and masters in bioprocess technology from ICT Mumbai and has obtained his Ph.D. in Chemical Engineering from IIT Delhi. Nikhil has 3 years' experience of working in R&D centres of leading Indian biotech companies like Intas and Syngene. He is a bioprocess engineer by training and is actively engaged in developing technology solutions for affordable manufacturing of biomolecules. Nikhil has authored more than 10 research publications in international peer reviewed journals, 2 book chapters and 2 patents and has presented at many international conferences.



KHARE RAVINDRA is the Co-founder and CEO of Symphony Technologies Pvt Ltd. He is an internationally acknowledged expert in Design, Manufacturing and Quality and applying statistical analytics to Engineering and BioPharma applications. He has published work on Data Analytics and Risk Management. He holds a master's degree in data science & engineering from BITS Pilani. He serves as an Emeritus Professor at the COEP Technological University, Pune.



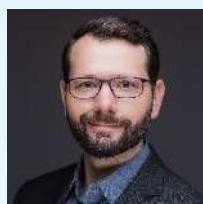
KULKARNI PRIYA is a Healthcare industry leader with 20+ years of global experience in drug development, bioprocess optimization, and digital health technologies (DHTs). Her expertise spans biostatistics, real-world evidence, and regulatory science across the U.S., EU, and emerging markets. At Genentech, Priya partnered with research scientists in bioprocess development, manufacturing, and quality control, leading statistical activities such as exploratory data analysis, immunogenicity assay validation, comparability studies, and applying Quality by Design (QbD) principles to the characterization of Nutropin, Pulmozyme, and Ocrelizumab. A dedicated educator, she has designed and delivered training programs for scientists, clinicians, and engineers on Design of Experiments, Principles of Biostatistics, and Data Visualization, enhancing statistical literacy and enabling data-driven decision-making at scale. She has contributed to industry innovation through TransCelerate Biopharma, the Digital Therapeutics Alliance, and Roche's Startup Creosphere, and has held roles at Merck, Genentech, Roche, and Pfizer. Priya holds a Ph.D. in Statistics (UC Santa Barbara) and an M.Sc. in Applied Statistics & Informatics (IIT Bombay). She recently relocated to Pune, India.



KUMAR SUNIL is currently a Postdoctoral Scientist at the Department of Drug Sciences, University of Pavia, Italy, where his research focuses on elucidating glycosylation patterns and applying glycoengineering strategies to monoclonal and polyclonal antibodies for vaccine development. Prior to joining the University of Pavia, he served as a Senior Postdoctoral Fellow in the Department of Chemical Engineering at the Indian Institute of Technology Delhi, India (2019–2024). In 2015, he was awarded a Start-up Research Grant under the prestigious Young Scientist Scheme from the Department of Science and Technology, India. Dr. Kumar earned his Ph.D. in Biotechnology from the University of Mysore, India, in 2013. He is internationally recognized for pioneering innovative 2D-LC-MS and Multi-Attribute Method (MAM) workflows to monitor critical quality attributes of monoclonal antibodies. Dr. Kumar integrates diverse chromatographic techniques with high-resolution mass spectrometry (Orbitrap and Q-TOF) and native MS to deliver regulatory-compliant, high-impact analytical insights. He has authored more than 25 publications in leading journals, including Analytical Chemistry, Talanta, and JASMS, and is frequently invited to present his work at international scientific meetings.



LEE E. K. is a Professor Emeritus at the Department of Bionano Engineering of Hanyang University-ERICA, Korea. He obtained his B.S. degree from Hanyang University, Korea in 1975 and a Ph.D. from Drexel University, Philadelphia, PA, USA in 1985, both in Chemical Engineering. From January 1985 through July 1992, he worked in the US for Miles Laboratories/Bayer and Pitman-Moore/Amgen in the field of recombinant proteins processing technology and protein structure-function analyses. Since his return to Korea in 1992 as a Professor of Chemical Engineering at Hanyang University-ERICA, he has been active in the R&D activities focusing on applying biochemical engineering principles to biopharmaceutical protein development and protein interactions analyses. He is a member of National Academy of Engineering in Korea and has served as a scientific and technical consultant to the Korean government and several biopharmaceutical industries globally. After the retirement in March 2018, he co-founded a bio startup, ImmunoForge, Ltd., and works as a VP and CQO leading the MSAT/CMC team to develop long-acting therapeutic biobetters as orphan drugs. Furthermore, he consults biotech companies on therapeutic antibody CDMO establishment and agarose-based chromatography resin development.



LINGG NICO is Senior Scientist and Head of Pharmaceutical Research at the Austrian Centre of Industrial Biotechnology (acib) and Senior Lecturer at BOKU University, where he leads a research lab on downstream processing of recombinant peptides, proteins, and bio nanoparticles for therapeutic applications, including vaccines. They serve as a European expert for the European Medicines Agency and on the editorial board of Vaccine. With 45 publications and book chapters, several patents, and leadership in international conferences such as ISPPP 2023, Nico Lingg contributes to advancing bioprocess science and engineering.



MAHARANA CHINMOYEE received her Ph.D from the National Brain Research Centre, Manesar, a nodal center under the Department of Biotechnology, India. Her doctoral research work was published in reputable international journals. Dr Chinmoyee did her 1st post-doctoral work with Prof. Mark P. Mattson, at the National Institute of Aging, NIH, USA. Dr. Chinmoyee worked both in basic and translational research to understand the roles of Presenilin 1 in Alzheimer's disease, and she was recognized by NIH, USA, and awarded the Fellow Award for Research Excellence 2015. Dr. Chinmoyee worked at the Indian Institute of Technology Delhi as a research scientist in the biotherapeutics and anti-venom project. She also serves as an editorial board member in international peer-reviewed journals and has published her research work and book chapters in reputed national and international peer-reviewed journals. She is now serving as an Assistant Professor, Department of Zoology, University of Jammu, and Deputy Director, Skill Innovation Incubation Entrepreneurship Development Centre and Science Club, Coordinator, and many of the institutional committees.



MATTEI AIMEE is the Director of Immunoinformatic at EpiVax. Her team provides silico immunogenicity risk assessments supporting research, collaborative, and commercial projects. She leads the development of novel in silico methods for assessing the immunogenic risk of biotherapeutics and their impurities. She holds an M.S. in Pharmaceutical Chemistry and has previous experience in the manufacturing of peptides, supporting research programs from concept assessment through lead optimization in the endocrinology and oncology therapeutic areas.



MISHRA AVINASH, established himself in the field of computational biology, holding a PhD from the Indian Institute of Technology Delhi (IITD), where he specialized in protein structure prediction. Prior to his doctoral studies, Avinash contributed to the research and development wing of AstraZeneca, where he honed his skills and knowledge in the pharmaceutical industry. His academic and research pursuits took a significant leap forward during his postdoctoral fellowship, where he was honored with the Indo- Australian Gold Fellowship. This prestigious award enabled him to conduct two years of research at Griffith University in Australia, further solidifying his expertise and contributions to the field. Avinash (CEO, Growdea Tech) and his team are at the forefront of computational biology with a focus on drug discovery. They are dedicated to empowering experimental biologists through computational methods, aiming to enhance the quality and efficiency of research. Avinash has been instrumental in designing and developing an in-silico drug discovery platform. This comprehensive package aims to accelerate the drug discovery process using efficient algorithms that incorporate machine learning and artificial intelligence. The team is now transitioning their empirical methods to AI based learning techniques, striving to improve the accuracy of predictions and make significant strides in the drug discovery domain.



MULE ABHISHEK holds a doctorate in Microbiology from the University of Mumbai, India. He joined Eppendorf Asia Pacific China in 2015 as a Senior In-field Application Specialist, where he provides application support, troubleshooting, and both theoretical and practical training in fermentation, scale-up, and bioreactor optimization. Abhishek has extensive experience cultivating microorganisms and cells at scales ranging from a few milliliters to thousands of Liters. Prior to joining Eppendorf, Abhishek worked as a Research Scientist at ICT Mumbai, where he was actively involved in scaling aerobic and anaerobic cultures to thousand-liter bioreactors. He also completed postdoctoral research at IIT Bombay, focusing on using bioreactors for effluent treatment.



PRAKASH OM is an assistant professor in the Department of Chemical Engineering at IIT Delhi. His research focuses on process systems engineering, where he utilizes data-driven and machine learning approaches to address various challenges. He earned his Ph.D. in Chemical Engineering from IIT Bombay, where he utilizes graph theory and developed optimization-based strategies for sensor placement for fault detection, diagnosis, and variable estimation for process systems, including water networks. Before joining IIT Delhi, he was a postdoctoral fellow at the University of Alberta, where he worked on data-driven modeling and estimation, anomaly detection, and soft sensor development, in collaboration with the oil and gas industry in Canada. His current interests span machine learning for process analytics, optimization, and the integration of AI/ML tools for decision support in chemical and biological systems.



P. KARUNDEV is a researcher in the Continuous Biopharma Manufacturing team at TCS Research, with expertise in bioreactors, integrated bioprocess simulations, Acoustic Wave Separator (AWS), and chromatography operations. Contributor to multiple patents and papers on continuous biopharma manufacturing.



PATANKAR DHANANJAY is an independent biopharmaceutical professional and freelance consultant, and is associated with several companies, startups and academic institutions in advisory roles. Previously he was the head of the biologics divisions of Syngene and Intas Pharmaceuticals and was senior scientist at Wockhardt. He has been intimately involved in the growth of the Indian biopharmaceutical industry since its early days. Over his career, he led teams that developed India's first biosimilar therapeutic product (EPO), India's first biosimilar approved for marketing in Europe (Filgrastim), India's first EU-GMP certified biologics manufacturing facility (Intas), and India's first commercial contract manufacturing of a novel biologic for the US market (Syngene). He has served in various national committees and biotechnology industry bodies in India and was Biologics Expert Committee member at the US Pharmacopeia from 2011 till 2020. By education he is a Chemical Engineer with bachelor's degree from IIT Mumbai and Ph.D. from University of Utah in the US.



PENNINGS MAARTEN received his M.Sc. in Bioprocess Technology from Wageningen University in The Netherlands and has 25 years of experience in the development of biopharmaceuticals. He spent 12 years at Xendo (now ProPharma Group) as a consultant process development, worked in projects on antibiotics, antibodies and viral vaccines. He has also been working on the development of continuous manufacturing in biopharma since 1999 and is a co-inventor of the BioSMB®, nowadays a Sartorius technology. For the past 10 years he has been in charge of process development and manufacturing for BiosanaPharma, a company developing biosimilar antibodies produced with a continuous manufacturing platform from perfusion bioreactor to bulk drug substance formulation and is now continuing these activities at Ardent Biopharma.



PRASAD SAI is the CTO for IoT and Digital Engineering at TCS based in Bangalore. Sai is a bachelor's in engineering from Bangalore University and a PG Diploma in Design Thinking & Innovation from Emeritus, Singapore. Sai has over 30 years of experience in working with various technologies in Digital Engineering and IoT. He is also skilled in product development and product management and has built products in different domains such as Life Sciences, Retail and Manufacturing. He has 4 granted patents related to Digital Twins for logistics, manufacturing, and process industries. His areas of interests include Design Thinking and fostering an innovation culture. He is also a registered Patent Agent at the Indian Patent Office



RAMTEKE MANOJKUMAR is Professor at the Department of Chemical Engineering, Indian Institute of Technology, Delhi, India. He is also Associate Head of CEP/QIP and Associate Faculty of Yardi School of Artificial Intelligence. Previously, he worked as a Scientist 1 at ASTAR Singapore and Assistant Professor in the Department of Polymer and Process Engineering, IIT Roorkee. His research interests include Modeling, Multi-Objective Optimization, Machine Learning, Process Sustainability, Ecosystem Services, Process Planning, and Scheduling. He has published more than 70 research papers in international journals, 6 book chapters, and several conference papers in these areas. He has also co-authored a textbook on optimization.



RATHORE ANURAG S. is an Institute Chair Professor at the Department of Chemical Engineering, Indian Institute of Technology, Delhi, India. He is also the Coordinator for the DBT COE for Biopharmaceutical Technology. His previous roles included management positions at Amgen Inc., Thousand Oaks, California and Pharmacia Corp., St. Louis, Missouri. He is presently serving as the Editor in Chief for The AAPS Journal. His areas of interest include process development, scale-up, technology transfer, process validation, biosimilars, continuous processing, formulation, biopharmaceutics, PAT and QbD. He has authored more than 800 publications and presentations in these areas. Prof. Rathore has been a recipient of the Tata Transformation Prize 2023 and Agilent Thought Leader Award 2020. He is presently serving as the Editor-in-Chief of *The AAPS Journal* and Associate Editor for *Journal of Chemical Technology and Biotechnology*. He also serves on the Editorial Advisory Boards for *Biotechnology Progress*, *Biotechnology Journal*, *Electrophoresis*, *BioPharm International*, *Journal of Chromatography B*, *Journal of Chromatography Open*, *Pharmaceutical Technology Europe*, and *Separation and Purification Reviews*. Dr. Rathore has edited books titled *Process Validation* (2023), *Preparative Chromatography for Separation of Proteins and Peptides* (2017), *Quality by Design for Biopharmaceuticals: Perspectives and Case Studies* (2009), *Elements of Biopharmaceutical Production* (2007), *Process Validation* (2005), *Electrokinetic Phenomena* (2004), and *Scale-up and Optimization in Preparative Chromatography* (2003). He has a Ph.D. in Chemical Engineering from Yale University.



RICHARD GUILHEM is the Chief Technology Officer of EpiVax, Inc. and has over 10 years of experience in computational immunology and vaccinology. Dr. Richard joined the EpiVax immunoinformatic team in 2014, developing new models for understanding the interaction between the immune system and protein sequences, while supporting EpiVax's commercial and research programs. He spearheaded the development of a new computational platform for the development of personalized, neoantigen-based cancer vaccines (Ancer), and conducted subsequent translational work at EpiVax's subsidiary company, EpiVax Therapeutics, Inc., from 2019 to 2024 where he served as Chief Scientific Officer/Chief Technology Officer. Dr. Richard has been leading innovative efforts at EpiVax since 2024 and is overseeing the development of its next generation of immunoinformatic tools. Dr. Richard holds a Ph.D. in Bioinformatics from Boston University and a M.Eng. in Bioinformatics and Modeling from the National Institute of Applied Sciences (INSA) of Lyon, France.



RUNKANA VENKAT is currently the Chief Scientist and Head of Engineering Sciences Research in TCS. Venkat is a chemical engineer by education and holds a Ph.D. in Environmental Engineering from Columbia University, New York. He has more than 34 years of experience in process modeling, simulation and optimization, advanced data analytics and digital twins, process development, scale-up and design, nanomaterials, batteries, carbon capture and biopharmaceutical manufacturing. Venkat is a Fellow of Indian National Academy of Engineering. He was an AICTE-INAE Distinguished Visiting Professor at IIT Kanpur during 2013-2018. Venkat received the TCS Distinguished Scientist Award in 2014 and his team received the Tata Group level innovation award, Tata Innovista for Implemented Innovations and Piloted Technologies, Aegis Graham Bell Award for Innovations in Manufacturing, and the IT Innovation Award from Express IT Awards for their work on digital twin technology. He is a co-inventor of 50+ granted and 100+ filed unique patents. He has also co-authored 12 chapters in edited volumes, 125 journal/conference papers and more than 150 conference presentations. Venkat's team has developed the software platform TCS INTWIN® for development and deployment of industrial digital twins and several other software assets and contributed to the development of TCS IP2™ (TCS Intelligent Power Plant) that uses artificial intelligence, IoT, and digital twin technologies to optimize the performance of power plants.



SABDE SUDEEP holds Ph.D. in Pharmaceutical Sciences and certification in General Management from IIM Kozhikode. He has over 16 years of experience in biosimilar development, small molecules, novel biologics drug discovery. His expertise includes Bioassays, PK Immunogenicity, cell biology and virology. A recipient of DBT and Lady Tata Memorial Trust fellowships, he has published 14 research articles, holds two international patents for a novel neutralizing antibody detection assay.

Dr. Sabde has contributed to R&D at leading pharmaceutical companies including Wockhardt, Lupin, Biocon Biologics, Nektar Therapeutics, and Syngene International Ltd

Dr. Sudeep Sabde currently serves as Senior Scientific Affairs Manager at USP, leading Scientific Affairs activities for USP Biologics in South Asia.



SHAH SANJAY B is a recognized SME in process engineering, automation, technical operations, strategic infrastructure planning, capital projects, and technology transfer. His expertise spans the entire biopharma lifecycle, from facility design and startup to GMP commercial operations and regulatory compliance. Sanjay has held leadership roles at globally renowned organizations, including Eli Lilly, Pfizer, Amgen, Dr. Reddy's Labs, Syngene, Intas, Kvaerner, and Hyde. In the US for 16 years, he served as Senior Process Engineer at Kvaerner, supporting diverse projects, and as Director of Process Engineering at Eli Lilly, supporting large-scale operations. He was lead downstream engineer for Amgen's Enbrel® project. After relocating to India in 2010, Sanjay established a cross-functional department at Dr. Reddy's Labs as Director, led CDMO operations at Syngene for global clients, and served as Vice President and Head of Manufacturing Operations & Site Engineering at Intas for biosimilar operations. As Vice President of Global Engineering at Pfizer, he supported the Asia-Pacific sterile injectables network. Most recently, he was VP of Global Engineering at Hyde, a US-based consulting firm. Holding an MS in Chemical Engineering from US, Sanjay is an active member of ISPE and PDA. His global expertise and leadership position him as a thought leader driving transformation and innovation in India's rapidly evolving biopharmaceutical landscape.



VIJAYAN VISHAL holds an MSc in Biotechnology from Goa University. He began his career at Serum Institute of India, working on Multiproduct Manufacturing of Recombinant vaccines, followed by a role at Merck Life Science as a Process Development Specialist (MSAT). Currently, he serves as Bioprocess Application Specialist at 3M India.



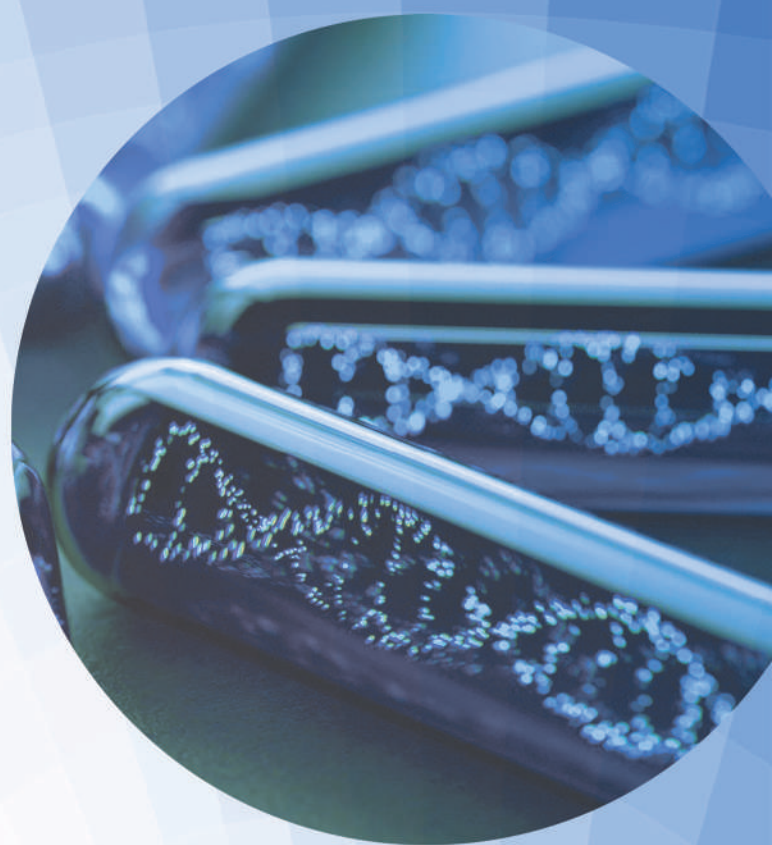
YEZHUVATH VINESH BALAKRISHNAN is part of the Strategic Capability Group within Life Sciences and Healthcare and is enabling customers in their business and IT transformation journey in areas of Manufacturing, Supply Chain, Quality, and Regulatory. through transformation, theme-based point of view, consulting, building assets, and driving business & growth. Vinesh brings over 33 years of experience in the pharmaceutical industry, including large-scale transformation programs, value engineering, digital transformation, advisory services, and innovation. Passionate and knowledgeable associate who is always eager to learn and contribute. combines process orientation and analytical abilities with an in-depth understanding of technology to develop IT solutions that drive productivity, efficiency, and digital maturity. He leads several strategic initiatives for Life Sciences, including Continuous Biopharma and Accelerating Regulatory Outcomes. Quality 5.0 and AI/GenAI/Agent AI.



YOON SEONGKYU is a Professor in the Francis College of Engineering at the University of Massachusetts (UMass), Lowell which he joined in 2010 after 15 years in industry. Currently, Dr. Yoon is co-director of Massachusetts Biomanufacturing Center, the UMass site director of the NSF/IUCRC/AMBIC (Advanced Mammalian Biomanufacturing Innovation Center), and the co-PI of NSF/AccelNet/IbioNe (International Biomanufacturing Network), Dr. Yoon founded the Massachusetts-based Biomanufacturing Institute. Yoon is leading a research group of systems and synthetic biology while conducting research in platform process development for protein, gene and vaccine therapeutics. He completed his PhD at McMaster University (Hamilton, Canada), and MBA from Babson College (Wellesley, MA).

Meet Our Team





Dec 2025
CBT COURSE SERIES



☎ +91-11-26596242

🌐 <https://cbt.iitd.ac.in>

✉ coe.biopharma.course@gmail.com